ICT Course: Introduction to Cryptography

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Session 5: Asymmetric Cryptography - Introductions

1. Introduction to Public Key Cryptography
   - Symmetric cryptography issues
   - Principle of asymmetric cryptography
   - Security mechanisms of Public Key Cryptography

Symmetric cryptography issues

- Key Distribution: The key must be established between Alice and Bob in secured channel
- Number of keys: a key for each pairs of users
- No protection against cheating of either Alice or Bob

Principle of asymmetric cryptography

- Trap door one-way function
- A key pair:
  - public key k_1: is broadcast to anyone
  - private key k_2 :is kept secret
- Algorithm: plaintext M, Ciphertext C

$$C = e_{k_1}(M)$$

$$M = e_{k_2}(C)$$

Security mechanisms of Public Key Cryptography

- Key Distribution without a pre-shared secret key, e.g, Diffie-Hellman key exchange, RSA
- Digital Signatures
  - providing integrity
  - preventing 'reputation'
  - idetification: using challenge-response protocols
- Encryption, e.g, RSA

Practical protocols

Most protocol are hybrid protocol, incorporate both symmetric and public-key algorithms:

- Key exchange and digital signature: use asymmetric algorithms (slow)
- Encryption: uses symmetric ciphers (fast)

How to build Public-Key algorithm

Asymmetric schemes are based on **one-way function** $f()$

- Computing $y = f(x)$: easy
- Computing $x = f^{-1}(y)$: infeasible

One-way function are based on mathematically hard problems:

- Factoring integers: given a composite integer n, find its prime factors
- Discrete Logarithm: given a, y, and m, find x such that $a^x = y \ mod \ m$
- Elliptic Curves: Generalization of discrete logarithm